

Computational modeling of neural populations

Context

Model types

Biological models: animal models (in-vivo), tissue models (ex-vivo), cell models (in-vitro)...



flies



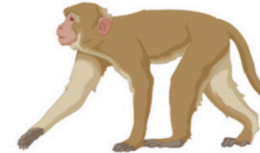
worms



mice



rats



monkeys

Computational models: phenomenological (minimal/toy models), biophysical...

$$\frac{du}{dt} = f(u, t), \quad u(t)$$

ODEs

$$\frac{\partial u}{\partial t} = f(u, x, t), \quad u(x, t)$$

PDEs

$$\frac{\partial u}{\partial t} = f(u, x, t) + \int [\dots] dy, \quad \text{IDEs}$$

IDEs

ODEs/PDEs/IDEs + noise

SDEs

phenomenological

biophysical

- simple

- complex

- representative

- exact mechanisms

- approximate

- (more) precise

- minimum detail

- detailed

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Why computational models?



Experimental data

*in-vivo, in-vitro,
ex-vivo, clinical...*



Hypothesis



Model design



Experimental
validation



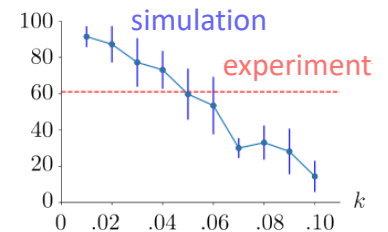
Prediction



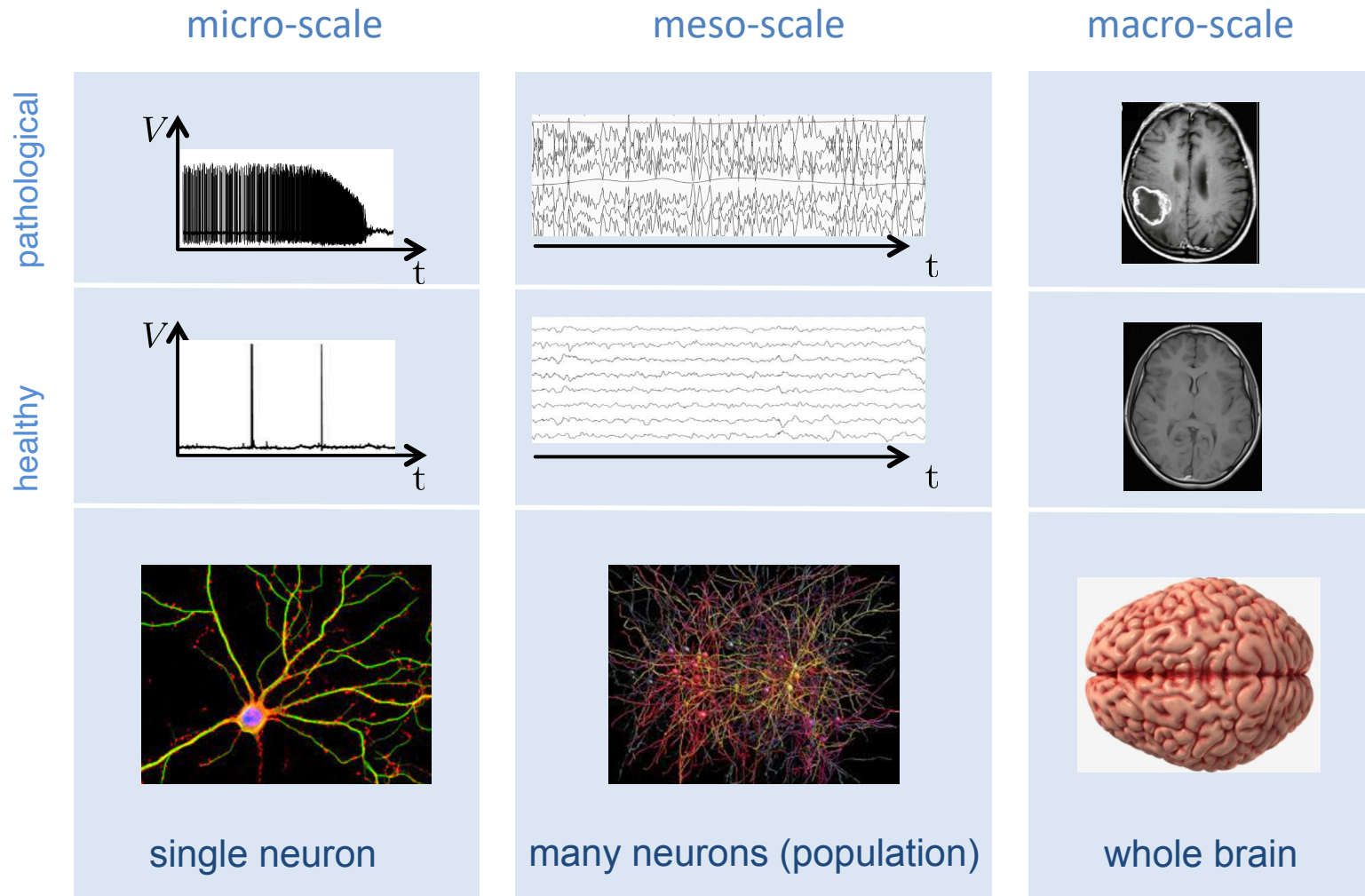
Simulation
analysis, fitting



Model
refinement



- to better understand the data
- to study *what-if* scenarios
- to make predictions (easier with precise parameter control)
- to facilitate the experiments



Chever et al.
J. of Clinical Inv. 2021

Lemaire et al.
PLOS C. Bio. 2021

Tschernatsch et al.
Int. Care Med. 2014

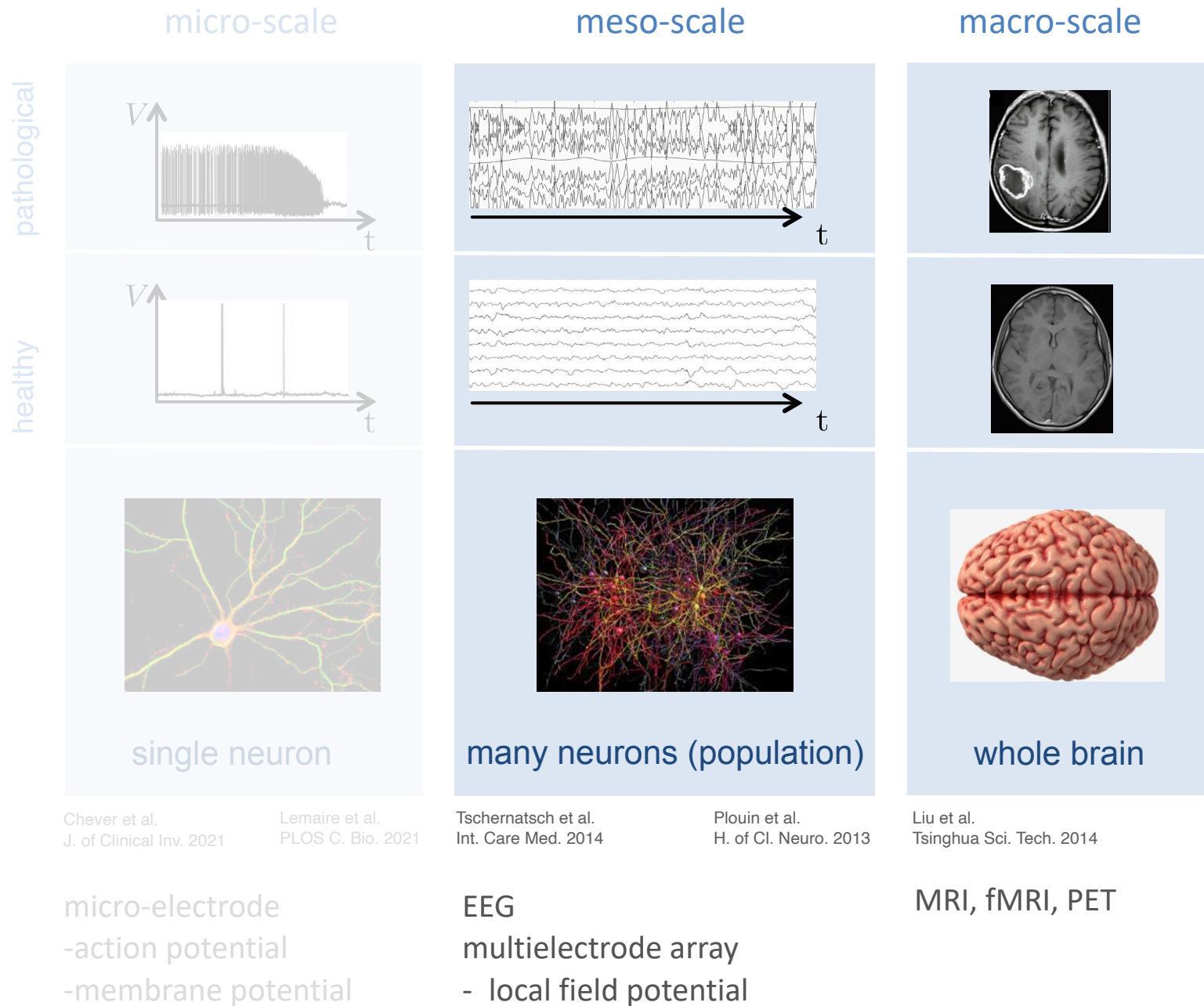
Plouin et al.
H. of Cl. Neuro. 2013

Liu et al.
Tsinghua Sci. Tech. 2014

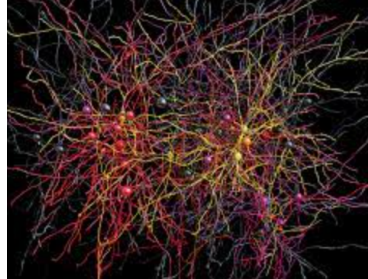
micro-electrode
-action potential
-membrane potential

EEG
multielectrode array
- local field potential

MRI, fMRI, PET



Modeling neural populations



Plouin et al.
H. of Cl. Neuro. 2013

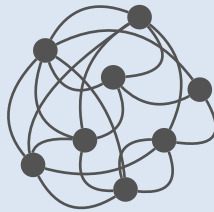
Networks

$$\frac{du_1}{dt} = f_1(u_1, \dots, u_n, t)$$

$$\frac{du_2}{dt} = f_2(u_2, \dots, u_n, t)$$

•
•
•

$$\frac{du_n}{dt} = f_n(u_1, \dots, u_n, t)$$

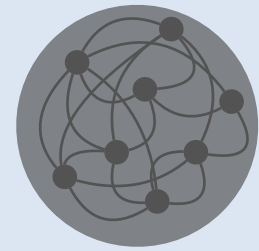


+ biophysically detailed
+ precise dynamics

- high dimensional
- difficult to manipulate
- computationally heavy

Low-dimensional approx.

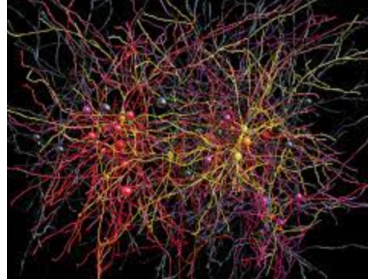
$$\frac{dU}{dt} = F(U, t)$$



+ low dimensional
+ simple
+ computationally light

- not detailed
- approximate dynamics

Modeling neural populations



Plouin et al.
H. of Cl. Neuro. 2013

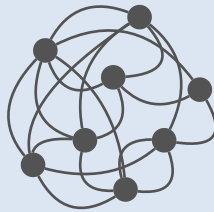
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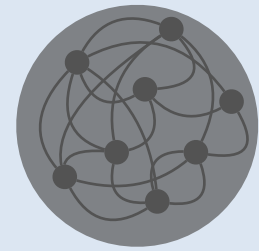
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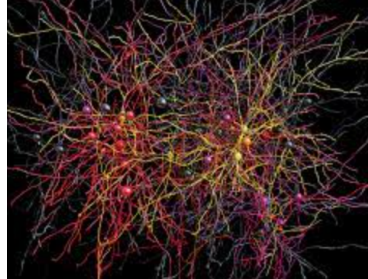
averaged dynamics



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Modeling neural populations



Plouin et al.
H. of Cl. Neuro. 2013

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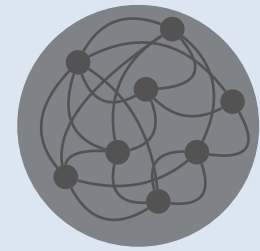
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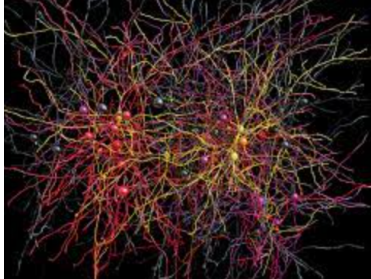
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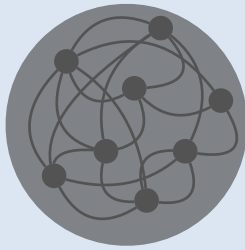
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- approximate dynamics

Modeling neural populations: applications



Plouin et al.
H. of Cl. Neuro. 2013

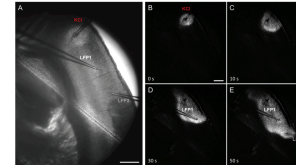
Low-dimensional approx.



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Neural activity pattern

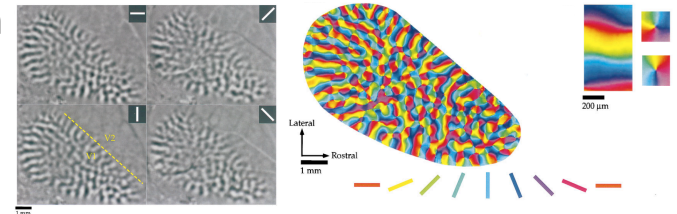
Periodic spiking
Propagating waves



Baspinar et al.,
PLOS CB, 2025

Visual perception

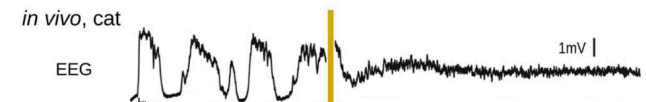
Orientation detection



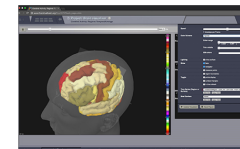
Bosking et al.,
J Neuroscience, 1997

Brain states

Awake
Sleep/anesthesia



Goldman et al., Front. Sys.
Neuroscience, 2019



- To get acquainted with computational modeling in neuroscience
- To acquire general knowledge about
 - neural fields
 - neurogeometry
 - The Virtual Brain (TVB)
- To acquire the ability to simulate these models for specific applications

Part I: Neural fields

Wilson-Cowan model
Spatially extended Wilson-Cowan model
Amari model

Application

Neural population dynamics

Part II: Neurogeometry

Petitot-Citti-Sarti approach

Application

Orientation detection

Part III: Whole-brain dynamics

The Virtual Brain (TVB)

Application

Brain states (awake, sleep/anesthesia)

The end